Rohit Kannan, Ph.D.

Assistant Professor, Operations Research

Curriculum Vitae

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Google Scholar
LOTUS Lab

EDUCATION

Ph.D.	Chemical Engineering	Massachusetts Institute of Technology	2018
M.S.	Chemical Engineering Practice	Massachusetts Institute of Technology	2014
B.Tech.	Chemical Engineering	Indian Institute of Technology Madras	2012

PROFESSIONAL APPOINTMENTS

Assistant Professor, Grado Department of Industrial and Systems Engineering, Virginia Tech	since Fall 2023
Faculty Researcher Guest, Los Alamos National Laboratory	2023–2025
Postdoctoral Associate, Applied Mathematics, Los Alamos National Laboratory	2021–2023
Postdoctoral Associate, Wisconsin Institute for Discovery, University of Wisconsin-Madison	2018–2020
Engineering Consultant for Alcon and Corning Incorporated	2013
Research Intern, Laboratory of Computational Systems Biotechnology, EPFL, Switzerland	2011
Research Intern, Technology Transfer Lab, Orchid Chemicals and Pharmaceuticals, India	2010

RESEARCH INTERESTS

Methodologies: Optimization under uncertainty, Global optimization, Machine learning-guided optimization, Computational optimization algorithms.

Applications: Energy systems, Process systems engineering.

SELECTED HONORS AND AWARDS

NSF CAREER Award	2026
Elected Vice-Chair of Global Optimization, INFORMS Optimization Society	2025–2026
INFORMS Senior Member (in recognition of sustained contributions to the community)	2025
Excellence in Teaching Spotlight, Center for Excellence in Teaching & Learning, Virginia Tech	2024
VT Principles of Community Group Award (as part of the ISE InclusiveVT Committee)	2024
Received 15 "Thank-a-Teacher notes" from students at Virginia Tech	since 2023
Center for Nonlinear Studies Postdoctoral Fellowship, Los Alamos National Laboratory	2021–2022
Best Paper Award, Advances in Optimization, 2015 AIChE Annual Meeting	2015
George M. Keller Graduate Fellowship, MIT	2012–2013
Oil and Natural Gas Corporation Scholarship, Government of India	2012
Reliance Heat Transfer Award and Institute Merit Awards for academic excellence, IIT Madras	2010–2012
EPFL, DAAD WISE, and MITACS Globalink Summer Research Fellowships (chose EPFL)	2011

FUNDING HISTORY

At Virginia Tech

Principal Investigator, "*CAREER: Foundations of Machine Learning-Enhanced Branch-and-Bound Algorithms for Scalable Global Optimization*," Operations Engineering Program, Division of Civil, Mechanical, and Manufacturing Innovation, National Science Foundation. 2026–2031
Total Amount: $639,433.

Principal Investigator, "*Iterative Design of Metal-Organic Frameworks Using Surrogate-Based Global Optimization*," Institute for Critical Technology and Applied Science (ICTAS) Engineering Faculty Organization-Opportunity (EFO-O) Seed Grant Program, Virginia Tech. 2024–2025
Co-PI: Dr. Luke Achenie. Total Amount: $10,000. Personal Share: $10,000.

Principal Investigator, New Faculty Mentoring Grant Program, Virginia Tech 2024–2026
Total Amount: $1,500.

Prior to Virginia Tech

Principal Investigator, "*Graph Convolutional Neural Networks to Accelerate Solutions to Hard Nonconvex Optimization Problems*," Applied Machine Learning Summer Research Fellowship Program, Information Science and Technology Institute, Los Alamos National Laboratory. 2023
Co-PIs: Dr. Harsha Nagarajan and Dr. Deepjyoti Deka. Total Amount: $25,000.
(Used to fund Erin George's summer internship at LANL.)

Co-Investigator, "*Stochastic Optimization Models for Enhancing the Resilience of the Power Grid*," Center for Nonlinear Studies Summer Student Program, Los Alamos National Laboratory. 2022
PIs: Dr. Harsha Nagarajan and Dr. Deepjyoti Deka. Total Amount: $10,000.
(Used to fund 50% of Mithun Goutham's summer internship at LANL.)

Co-Investigator, "*Learning to Accelerate Global Solutions for Non-Convex Optimization*," Laboratory Directed Research & Development, Los Alamos National Laboratory. 2022–2025
PIs: Dr. Harsha Nagarajan and Dr. Deepjyoti Deka. Total Amount: $1,013,000.
(Used, in part, to fund 25% of my postdoctoral appointment at LANL.)

PUBLICATIONS

___ = Graduate Advisee, ⋆ = Undergraduate Advisee, ^{c} = Corresponding Author(s)

Peer-Reviewed Journal Papers

12. **R. Kannan**^{c}, H. Nagarajan^{c}, and D. Deka (2025), "Strong Partitioning and a Machine Learning Approximation for Accelerating the Global Optimization of Nonconvex QCQPs," *INFORMS Journal on Computing*, 38(5), pp. 1323–1343. [URL] [PDF]
11. E. M. Turan^{c}, J. Jäschke, and **R. Kannan**^{c} (2025), "Bounding-Focused Discretization Methods for the Global Optimization of Nonconvex Semi-Infinite Programs," *Computational Optimization and Applications* (special issue on global optimization algorithms and applications), 92(3), pp. 1035–1068. [URL] [PDF]
10. **R. Kannan**^{c}, G. Bayraksan, and J. R. Luedtke (2025), "Technical Note—Data-Driven Sample Average Approximation with Covariate Information," *Operations Research*, 73(6), pp. 3245–3259. [URL] [PDF]
9. **R. Kannan**^{c}, G. Bayraksan, and J. R. Luedtke (2024), "Residuals-Based Distributionally Robust Optimization with Covariate Information," *Mathematical Programming (Series A)*, 207(1–2), pp. 369–425. [URL] [PDF]
8. A. Subramanian^{c}, **R. Kannan**, F. Holtorf, T. A. Adams II, T. Gundersen, and P. I. Barton (2023), "Optimization Under Uncertainty of a Hybrid Waste Tire and Natural Gas Feedstock Flexible Polygeneration System Using a Decomposition Algorithm," *Energy*, 284, 129222, pp. 1–11. [OA URL]
7. **R. Kannan**^{c} and J. R. Luedtke (2021), "A Stochastic Approximation Method for Approximating the Efficient Frontier of Chance-Constrained Nonlinear Programs," *Mathematical Programming Computation*, 13(4), pp. 705–751. [URL] [PDF]
6. **R. Kannan**, J. R. Luedtke, and L. A. Roald^{c} (2020), "Stochastic DC Optimal Power Flow with Reserve Saturation," *Electric Power Systems Research* (special issue for the XXI *Power Systems Computation Conference*), 189, 106566, pp. 1–9. [URL] [PDF]. *Featured on the UW-Madison College of Engineering Website*. [URL]
5. **R. Kannan** and P. I. Barton^{c} (2018), "Convergence-Order Analysis of Branch-and-Bound Algorithms for Constrained Problems," *Journal of Global Optimization*, 71(4), pp. 753–813. [URL] [PDF]
4. **R. Kannan** and P. I. Barton^{c} (2017), "The Cluster Problem in Constrained Global Optimization," *Journal of Global Optimization*, 69(3), pp. 629–676. [URL] [PDF]
3. **R. Kannan** and A. K. Tangirala^{c} (2014), "Correntropy-Based Partial Directed Coherence for Testing Multivariate Granger Causality in Nonlinear Processes," *Physical Review E*, 89(6), 062144, pp. 1–15. [URL] [PDF]

Submitted Journal Papers

2. D. Gupta, N. Gotawala, R. Gnanasambandam, **R. Kannan**, H. Z. Yu, J. Yu, Z. Kong^{c}, "Temperature-Dependent Multiphysics Modeling of Additive Friction Stir Deposition Using Multi-Task Coupled Physics-Informed Neural Networks," arXiv preprint arXiv:2610.04082. Under Review. [PDF]

Technical Reports

1. **R. Kannan**[c], G. Bayraksan, and J. R. Luedtke (2021), "Heteroscedasticity-Aware Residuals-Based Contextual Stochastic Optimization," arXiv preprint arXiv:2101.03139. [PDF]

Working Papers

6. D. Sundararajan, J. Sanborn[★], G. Esperanza[★], and **R. Kannan**, "Machine Learning-Guided Undercover for Mixed-Integer Nonlinear Programs."
5. E. George, H. Nagarajan, D. Deka, and **R. Kannan**, "Learning to Accelerate the Global Optimization of QCQPs: An End-to-End Graph-Based Stochastic Partitioning Approach."
4. M. Goutham, J.-C. Raymond-Bertrand, D. Deka, H. Nagarajan, R. Bent, and **R. Kannan**, "Dispatch of Mobile Power Sources for Enhanced Grid Resilience using Ensemble Hurricane Forecasts."
3. J.-C. Raymond-Bertrand, N. Ho-Nguyen, J. R. Luedtke, and **R. Kannan**, "Data-Driven Multistage Stochastic Optimization on Time Series."
2. **R. Kannan**, A. Subramanian, and P. I. Barton, "GOSSIP: Decomposition Software for the Global Optimization of Nonconvex Two-Stage Stochastic Mixed-Integer Nonlinear Programs."
1. **R. Kannan** and P. I. Barton, "Integrating Benders Decomposition and Lagrangian Relaxation for Solving Two-Stage Stochastic Mixed-Integer Nonlinear Programs."

Peer-Reviewed Conference Papers

2. E. M. Turan, **R. Kannan**, and J. Jäschke[c] (2022), "Design of PID Controllers Using Semi-Infinite Programming," Proceedings of the 14th *International Symposium on Process Systems Engineering*, pp. 439–444. [URL] [PDF]
1. **R. Kannan** and P. I. Barton[c] (2016), "The Cluster Problem in Constrained Global Optimization," Proceedings of the 13th *Global Optimization Workshop* (GOW'16), pp. 9–12. [OA URL]

Theses

2. Ph.D.: "Algorithms, Analysis, and Software for the Global Optimization of Two-Stage Stochastic Programs," *Massachusetts Institute of Technology*, 2018. [URL] [PDF]
1. B.Tech.: "Partial Directed Coherence for Nonlinear Granger Causality: A Generalized Correlation Function-Based Approach," *Indian Institute of Technology Madras*, 2012.

TEACHING

At Virginia Tech

Instructor, Department of Industrial and Systems Engineering since Fall 2023

SPOT evaluations (max. value = 6) correspond to the question "Overall, the instructor's teaching was effective." Students submitted "Thank-a-Teacher notes" through the Center for Excellence in Teaching and Learning.

Term	Course	Enrollment	SPOT Mean, Median	# Thank You Notes
Sp26	ISE 6514: Advanced Math Programming	11	5.73, 6	2
Sp25	ISE 5406: Nonlinear Programming	15	5.46, 6	5
Fa24	ISE 3434: Deterministic Operations Research II	12	5.10, 5	2
Sp24	ISE 5406: Nonlinear Programming	13	4.64, 5	4
Fa23	ISE 6514: Advanced Math Programming	11	5.90, 6	2

Course Content

- ISE 3434 (Undergraduate Elective): complexity theory, integer programming, nonlinear programming, stochastic programming, machine learning, and modeling languages and optimization software.

- ISE 5406 (Graduate Core): convex analysis and optimization, generalizations of convexity, first and second-order optimality conditions, algorithms for unconstrained and constrained nonlinear optimization.
- ISE 6514 (Graduate Elective): convex analysis, optimality conditions, duality, LPs, decomposition methods, conic programming (SOCPs, SDPs), solution approaches, applications in OR, energy, ML.

Guest Lectures

➤ ISE 5024 (Graduate Seminar Series) 2023–2026

At MIT

Teaching Assistant, Department of Chemical Engineering 2015

- Teaching assistant for the graduate-level "Chemical Reactor Engineering" course (≈ 50 students).
- Designed homework exercises in MATLAB and Q-Chem, moderated online discussion forums.

STUDENT ADVISING AND MENTORING

Graduate Research Advising

Current Ph.D. Advisees

➤ Dhruva Sundararajan, since Fall 2024.

Grado Graduate Teaching Assistantship, ISE Department, Virginia Tech, 2024–2025.

Session Chair, "Machine Learning and Global Optimization," INFORMS Annual Meeting, Nov. 2026.

➤ Jean-Christophe ("Chris") Raymond-Bertrand, since Fall 2025.

Rehmert-Nachlas and Sweers Graduate Fellowships, ISE Department, Virginia Tech, 2025–2026.

Travel Award to attend the 2026 Artificial Intelligence School for Computer Science and Operations Research Education (AI-SCORE 2026)

Travel Award to attend the 2027 Grid Science Winter School and Conference (URL)

➤ Youngdoo Choi, since Fall 2026.

Grado and Sweers Graduate Fellowships, ISE Department, Virginia Tech, 2026–2027.

Current M.S. Advisees

➤ Atharva Shailesh Anchalwar, since Spring 2025. Co-advised with Dr. Luke Achenie in Chemical Engineering.

Former M.S. Advisees

➤ Padmapriya Rengasamy, M.S. in ISE, 2026. Co-advised with Dr. Luke Achenie in Chemical Engineering.
Thesis: "Inverse Screening of Metal-Organic Frameworks for Low-to-Mid-Pressure Carbon Dioxide Adsorption using Geometric, AP-RDF, and RAC Descriptors." [OA URL]

First Position: Digital Procurement Transformation Specialist, Colgate-Palmolive.

➤ Hiral Rajendrakumar Makwana, M.S. in ISE, 2025. Co-advised with Dr. Eduardo Molina in Sustainable Biomaterials.
Thesis: "Prediction modelling of pallet overhang on box compression strength." [OA URL]

Featured in Virginia Tech News, "Smarter Packaging Design", October 2025.

First Position: Ph.D. Student, Packaging Science, Department of Sustainable Biomaterials, Virginia Tech.

Undergraduate Research Mentoring

Current Undergraduate Researchers

➤ Alex Sobrevias Pueyo, since Fall 2026.

➤ Anannya Karthik, since Fall 2026.

- Elizabeth LaFalce, since Fall 2026.
- Gavinmarc Esperanza, since Spring 2025.

  NSF REU Research Fellow, Clemson University, Summer 2026.
- Jesus Alfaro, since Fall 2026.
- Nimah Rahman, since Fall 2026.

Former Undergraduate Researchers

- Jude Sanborn, Spring 2025 to Summer 2026.

  First Position: Ph.D. Student, Department of Industrial & Systems Engineering, UW-Madison.

  Outstanding Junior Award, ISE Department, Virginia Tech, 2025.

  Outstanding Academic Achievement Award, ISE Department, Virginia Tech, 2026.
- Jiaqi Wu, Spring 2026.

  First Position: M.S. Student, Industrial & Systems Engineering, Georgia Tech.
- Pedro Barcha, Spring 2026.

  First Position: M.S. Student, Data Analytics & Artificial Intelligence in Health Sciences, Bocconi University and Humanitas University.
- Yabsira Habtamu, Spring 2026. Continued as a junior in the VT ISE department.
- Antonio Ortega Vicente, Fall 2025. Continued as a sophomore in the VT ISE department.
- Soham Mehra, Spring 2024. Continued as a junior in the VT ISE department.

Senior Design Project Advising

Senior Design Advisor for the following student groups (15 students in total):

- Sean Borachok, Anthony Janay, Evan Mascia, and Cole Taylor, "Parks Division Mowing Operations Efficiency and Routing," 2026–2027.
- Hamdan Alzaabi, Sanjna Jotwani, and Ly Ritz, "Energy Performance Reporting Strategic Targets," 2025–2026.

  People's Choice Award, ISE Senior Design Symposium, Virginia Tech.
- Christopher Lee, Rithvik Mantha, Maguire McMahon, and Sydney Weeks, "LMI Resource Optimization Tool (ResOTo) version 2.0," 2024–2025.

  Honorable Mention, Process Improvement Track, 7th Andrew P. Sage Memorial Design Competition.
- Aidan Ford, Kareem Hamadeh, Finn Rooney, and Thomas Smiddy, "Amtrak Equipment Inventory Analysis," 2023–2024.

High School Research Mentoring

- Prateek Reddy, Summer 2026.

Ph.D. Students Mentored Prior to Virginia Tech

- Erin George (UCLA), "Graph Convolutional Neural Networks to Accelerate Solutions to Nonconvex Optimization," Applied Machine Learning Summer Research Fellow, *Los Alamos National Laboratory*, 2023.
  Role: Primary Mentor. Co-Mentors: Dr. Harsha Nagarajan and Dr. Deepjyoti Deka.
  First Position: NSF Postdoctoral Fellow, UC San Diego.

- Mithun Goutham (OSU), "Stochastic Optimization Models for Enhancing the Resilience of the Power Grid," Summer Student at the Center for Nonlinear Studies, *Los Alamos National Laboratory*, 2022.
  Role: Primary Mentor. Co-Mentors: Dr. Harsha Nagarajan and Dr. Deepjyoti Deka.
  First Position: Assistant Teaching Professor, Illinois Institute of Technology.
  Current Position: Teaching Assistant Professor, University of Washington.
- Avinash Subramanian (NTNU), "Optimization under Uncertainty of Hybrid Feedstock Polygeneration Systems," Ph.D. Student, Department of Energy & Process Engineering, *Norwegian University of Science and Technology*, 2021. Role: Mentor. Ph.D. Advisors: Dr. Truls Gundersen and Dr. Johannes Jäschke.
  First Position: Research Scientist, SINTEF, Norway. Current Position: Software Engineer, JuliaHub.

SERVICE

Professional Leadership and Service

- Elected Vice-Chair of Global Optimization, INFORMS Optimization Society 2025–2026
  - Cluster Chair, Global Optimization, 2026 INFORMS Annual Meeting
  - Track Chair, Global Optimization, 2026 INFORMS Optimization Society Conference
  - Cluster Chair, Global Optimization, 2025 INFORMS Annual Meeting
- Program Committee, Refereed Papers, INFORMS Optimization Society Conference 2026
- Poster Competition Committee, Mixed Integer Programming Workshop 2026
- Session Chair at the following conferences:
  - "Advances in Mixed-Integer and Global Optimization Under Uncertainty I & II," *INFORMS Annual Meeting* 2026
  - "Machine Learning for Global Optimization," *INFORMS Optimization Society Conference* 2026
  - "Heuristic and Hybrid Methods in Global Optimization," *INFORMS Optimization Society Conference* 2026
  - "Advances in Global Optimization: Algorithms and Applications," *INFORMS Optimization Society Conference* 2026
  - "Advances in Deterministic Global Optimization," *INFORMS Annual Meeting* 2025
  - "Recent Advances in Mixed-Integer and Global Optimization," *INFORMS Annual Meeting* 2025
  - "Contextual Stochastic Optimization" (Minisymposium and Invited Session), *International Conference on Stochastic Programming* 2025
  - "Mixed-Integer Programming-Based Techniques for the Global Optimization of MINLPs," *International Conference on Continuous Optimization* 2025
  - "Learning and Optimization Techniques for Uncertain Systems," *INFORMS Annual Meeting* 2024
  - "Advances in Deterministic Global Optimization," *INFORMS Optimization Society Conference* 2024
  - "Machine Learning for Discrete and Global Optimization I & II," *INFORMS Annual Meeting* 2023
  - "Learning-Assisted Dynamic Decision-Making Under Uncertainty," *INFORMS Annual Meeting* 2022
  - "Global Optimization of Stochastic and Semi-Infinite Programs," *INFORMS Annual Meeting* 2022
  - "Computational Stochastic Programming," *INFORMS Annual Meeting* 2018
- Organizer of monthly teleconference meetings on "Optimization Under Uncertainty" as part of the DOE MACSER Project (≈ 15 participants) 2018–2020

Peer Review

- Invited Peer-Reviewer for the following funding programs:
  - CAREER panelist for the U.S. National Science Foundation, 2026.
  - Ad hoc reviewer for the U.S. Air Force Office of Scientific Research, 2025.

- Ad hoc reviewer for the U.S. National Science Foundation, 2025.
- Panelist for the U.S. National Science Foundation, 2024.

➤ Invited Peer-Reviewer for the following journals and conferences (Web of Science Reviewer Profile):

- *Operations Research*
- *Mathematical Programming*
- *SIAM Journal on Optimization*
- *Journal of Global Optimization*
- *Management Science*
- *Mathematics of Operations Research*
- *Set-Valued and Variational Analysis*
- *Annals of Operations Research*
- *IISE Transactions*
- *Electric Power Systems Research*
- *Computers and Chemical Engineering*
- *Chemical Engineering Science*
- *Journal of Optimization Theory and Applications*
- *Computational Optimization and Applications*
- *INFORMS Journal on Computing*
- *Optimization Methods and Software*
- *Optimization Letters*
- *INFORMS Journal on Optimization*
- *Journal of Machine Learning Research*
- *IEEE Transactions on Control Systems Technology*
- *IEEE Transactions on Automation Science and Engineering*
- *American Control Conference*
- *Industrial and Engineering Chemistry Research*

University and Department Service

➤ Served in the following committees at Virginia Tech:

- Operations Research Faculty Search Committee, ISE Department 2025–2027
- Graduate Admissions Committee, ISE Department 2023–2027
- InclusiveVT Committee, ISE Department 2023–2025

➤ Invited Panelist, Awardee Panel, College of Engineering NSF CAREER Club, Virginia Tech 2026

➤ Invited Talk, "Challenge with Care: Designing for Rigor, Equity, and Engagement," Virtual Welcome Back Symposium on Practical Strategies to Enhance Student Learning, Center for Excellence in Teaching and Learning, Virginia Tech 2025

➤ Judge, ISE Graduate Research Poster Competition, Virginia Tech 2023, 2024

➤ Moderator, ISE Senior Design Symposium, Virginia Tech 2025, 2026

Graduate Committees and Examining

➤ Thesis Committee Member for the following Ph.D. students at VT (excluding my advisees):

- Minseo Lee (ISE), 2026–present. Advisor: Dr. Esra Büyüktahtakın Toy
- Hiral Rajendrakumar Makwana (SBIO), 2026–present. Advisor: Dr. Eduardo Molina
- Cinar Ari (ISE), 2026–present. Advisor: Dr. Robert Hildebrand
- Seung Jin Choi (ISE), 2026–present. Advisor: Dr. Esra Büyüktahtakın Toy
- Nipun Galande (ISE), 2026–present. Advisor: Dr. Esra Büyüktahtakın Toy
- Dhrubajyoti Gupta (ISE), 2025–present. Advisor: Dr. Zhenyu ("James") Kong.
- Asha Barua (ISE), 2025–present. Advisor: Dr. Sajad Khodadadian.
- Wasif Jawad Hussain (ECE), 2025–present. Advisors: Dr. Harpreet Dhillon and Dr. Mike Buehrer.
- Daniel A. Neira-González (ISE), 2023–present. Advisors: Dr. Subhash Sarin and Dr. Manish Bansal.
- Dr. Hyunwoo Lee (ISE), 2023–2026. Advisors: Dr. Esra Büyüktahtakın Toy and Dr. Robert Hildebrand.

➤ Invited External Examiner for the following students:

- Mari E. Rugland, "An object-oriented framework for the optimization of flexible renewable energy systems," M.Sc. thesis, Department of Chemical Engineering, *Norwegian University of Science and Technology* (NTNU), 2021. Advisors: Dr. Johannes Jäschke, Dr. Truls Gundersen, and Avinash Subramanian.

- Petter E. Nordby, "Optimization of flexible renewable energy systems under uncertainty," Master's thesis, Department of Energy & Process Engineering, *Norwegian University of Science and Technology* (NTNU), 2021. Advisors: Dr. Truls Gundersen, Dr. Johannes Jäschke, and Avinash Subramanian.

Educational Outreach

➤ Organized eight two-hour hands-on sessions titled "Crack the Code: An Interactive Journey Through Optimization" at Virginia Tech
- Watford Engineering Experience High School Program. [Slides] 2026
- C-Tech^2 High School Program. [Slides] 2025

➤ Served as a panelist for the following events:
- C-Tech^2 High School Program 2026
- TechGirls International High School Exchange Program 2025
- Watford Engineering Experience High School Program 2025
- Galipatia Slush Rush 2025
- Alpha Pi Mu Undergraduate Research Panel 2024

➤ Math Lecturer for the IIT Joint Entrance Exam (IIT JEE) 2016
- Recorded online video lectures for the entrance exam to the IITs as part of an MIT team
- Featured on MIT OpenCourseWare and supported by the MIT Office of Digital Learning

➤ Math Olympiad Trainer, Science and Math Academy for Real Talents, Chennai, India 2008–2011
- Coached $\approx$ 30 middle-school and high-school students each year for the Math Olympiad
- One trainee was among 40 students selected to represent India at the 2010 *International Math Olympiad Training Camp*

➤ Volunteer, National Services Scheme, IIT Madras 2008–2009
- Volunteered for the "Science Activities" team of National Services Scheme at IIT Madras
- Designed and demonstrated science experiments to students in underprivileged schools

PROFESSIONAL MEMBERSHIPS

➤ Senior Member, Institute for Operations Research and the Management Sciences (INFORMS). Also a member of INFORMS Optimization and Computing Societies

➤ Mathematical Optimization Society (MOS)

➤ Institute of Industrial and Systems Engineers (IISE)

➤ American Institute of Chemical Engineers (AIChE)

RESEARCH IN THE INDUSTRY

➤ Engineering Consultant, Alcon, Fort Worth, TX 2013
- Led an MIT team to design a measure of residual stress in molded intraocular lens wafers
- Increased bioavailability of an opthalmic drug through experiments, pharmacokinetic modeling

➤ Engineering Consultant, Corning Inc., Corning, NY & Wilmington, NC 2013
- Led an MIT team to model effects of oxygen on the cure rate of UV-curable acrylate polymers
- Accelerated cell media development by identifying key components using statistical analysis

➤ Research Intern, Orchid Chemicals and Pharmaceuticals, India 2010
- Designed and implemented solutions to scale up the production of a research drug
- Investigated methods for increasing the size of drug crystals to reduce filtration time

TECHNICAL SKILLS

- C++ & C
- Julia
- MATLAB
- Fortran
- Shell Scripting
- Python
- R
- GAMS
- Git
- Cluster Computing

PRESENTATIONS

Excludes presentations where I was a coauthor but not the speaker.

Area Plenary

2. "A Stochastic Approximation Method for Approximating the Efficient Frontier of Chance-Constrained Nonlinear Programs," CAST Division Plenary, *AIChE Annual Meeting*, Nov. 2021.
1. "GOSSIP: Decomposition Software for the Global Optimization of Nonconvex Two-Stage Stochastic Mixed-Integer Nonlinear Programs," CAST Division Plenary, *AIChE Annual Meeting*, Nov. 2016.

Invited

45. "Machine Learning-Guided Undercover for Mixed-Integer Nonlinear Programs," *Mixed Integer Programming Workshop*, May 2026.
44. "New Discretization Methods for the Global Optimization of Nonconvex Semi-Infinite Programs," *INFORMS Optimization Society Conference*, March 2026.
43. "Learning for Global Optimization," Faculty Meeting, Sanghani Center for Artificial Intelligence and Data Analytics, *Virginia Tech*, March 2026.
42. "Data-Driven Multistage Stochastic Optimization on Time Series," *Banff International Research Station Workshop on Contextual Stochastic Optimization*, Feb. 2026.
41. "Learning to Accelerate the Global Optimization of QCQPs: Strong Partitioning and an End-to-End Graph ML-Based Approach," *INFORMS Annual Meeting*, Oct. 2025.
40. "Learning to Accelerate the Global Optimization of QCQPs: Strong Partitioning and an End-to-End Graph ML-Based Approach," *8th International Conference on Continuous Optimization (ICCOPT)*, July 2025.
39. "Data-Driven Multistage Stochastic Optimization on Time Series," *INFORMS Annual Meeting*, Oct. 2024.
38. "Learning to Accelerate the Global Optimization of Quadratically-Constrained Quadratic Programs," *25th International Symposium on Mathematical Programming*, July 2024.
37. "Bounding-Focused Discretization Methods for the Global Optimization of Nonconvex Semi-Infinite Programs," *INFORMS Optimization Society Conference*, March 2024.
36. "LE^{2}GO: Learning An End-to-End Partitioning Policy for the Global Optimization Of QCQPs," *INFORMS Annual Meeting*, Oct. 2023.
35. "Data-Driven Multistage Stochastic Optimization on Time Series," *16th International Conference Stochastic Programming (ICSP)*, July 2023.
34. "Learning to Accelerate the Global Optimization of Quadratically-Constrained Quadratic Programs," *SIAM Conference on Optimization*, May 2023.
33. "Learning-Assisted Data-Driven Optimization," Department of Chemical and Biological Engineering, *South Dakota School of Mines and Technology*, March 2023.
32. "Learning-Assisted Data-Driven Optimization," Grado Department of Industrial and Systems Engineering, *Virginia Tech*, Feb. 2023.
31. "Integrating Time Series Predictions Within Multistage Stochastic Optimization," *INFORMS Annual Meeting*, Oct. 2022.
30. "Data-Driven Multistage Stochastic Optimization on Time Series," *7th International Conference on Continuous Optimization (ICCOPT)*, July 2022.
29. "Learning-Assisted Data-Driven Optimization Under Uncertainty," School of Mathematics, *University of Edinburgh*, April 2022.

28. "Learning-Assisted Data-Driven Optimization Under Uncertainty," Department of Mechanical and Manufacturing Engineering, *University of Calgary*, Feb. 2022.
27. "Learning-Assisted Data-Driven Optimization Under Uncertainty," School of Industrial Engineering and Management, *Oklahoma State University*, Feb. 2022.
26. "Residuals-Based Distributionally Robust Optimization with Covariate Information," *INFORMS Annual Meeting*, Oct. 2021.
25. "Stochastic DC Optimal Power Flow With Reserve Saturation," *Los Alamos–Arizona Days*, May 2021.
24. "A Modular Framework for Integrating Machine Learning within Stochastic Optimization," Department of Mathematical Sciences, *Florida Institute of Technology*, March 2021.
23. "A Modular Framework for Integrating Machine Learning within Stochastic Optimization," Department of Chemical and Biomolecular Engineering, *Clarkson University*, March 2021.
22. "A Modular Framework for Integrating Machine Learning within Stochastic Optimization," Systems and Industrial Engineering Department, *University of Arizona*, Feb. 2021.
21. "Data-Driven Sample Average Approximation with Covariate Information," *INFORMS Annual Meeting*, Nov. 2020.
20. "Stochastic DC Optimal Power Flow with Reserve Saturation," *INFORMS Annual Meeting*, Nov. 2020.
19. "Data-Driven Stochastic Optimization with Covariate Information," Mathematics & Computer Science Division, *Argonne National Laboratory*, Sep. 2020.
18. "Predict, then Smart Optimize with Stochastic Programming," Center for Nonlinear Studies, *Los Alamos National Laboratory*, July 2020.
17. "Predict, then Smart Optimize with Stochastic Programming," *IPAM Workshop on Intersections between Control, Learning and Optimization*, Feb. 2020.
16. "GOSSIP: Decomposition Software for the Global Optimization of Nonconvex Two-Stage Stochastic Mixed-Integer Nonlinear Programs," *INFORMS Annual Meeting*, Nov. 2018. Session: Computational Stochastic Programming.
15. "Algorithms, Analysis, and Software for the Global Optimization of Chemical Process Systems Under Uncertainty," *ExxonMobil Research and Engineering*, Clinton, NJ, Dec. 2017.
14. "Algorithms, Analysis, and Software for the Global Optimization of Chemical Process Systems Under Uncertainty," *Lawrence Berkeley National Laboratory*, Nov. 2017.
13. "Algorithms, Analysis, and Software for the Global Optimization of Chemical Process Systems Under Uncertainty," Wisconsin Institute for Discovery, *University of Wisconsin-Madison*, Oct. 2017.
12. "Optimization of Chemical Process Systems Under Uncertainty," *Rockwell Automation R&D*, Austin, TX, March 2017.
11. "Convergence-Order Analysis of Lower Bounding Schemes for Constrained Global Optimization Problems," *5th International Conference on Continuous Optimization (ICCOPT)*, Aug. 2016.

Contributed

10. "Learning to Accelerate the Global Solution of Quadratically-Constrained Quadratic Programs," *AIChE Annual Meeting*, Nov. 2022.
9. "Tighter Lower Bounds for Semi-Infinite Programming Using Parametric Sensitivity Theory," *AIChE Annual Meeting*, Nov. 2022.
8. "Data-Driven Multistage Stochastic Optimization on Time Series," *AIChE Annual Meeting*, Nov. 2021.
7. "Data-Driven Sample Average Approximation with Covariate Information," *AIChE Annual Meeting*, Nov. 2021.
6. "Residuals-Based Distributionally Robust Optimization with Covariate Information," *Robust Optimization Webinar*, April 2021.
5. "The Cluster Problem in Constrained Global Optimization," *AIChE Annual Meeting*, Nov. 2016.
4. "The Cluster Problem in Constrained Global Optimization," *13th Global Optimization Workshop*, Aug. 2016.

3. "Convergence-Order Analysis of Branch-and-Bound Algorithms for Constrained Problems," *AIChE Annual Meeting*, Nov. 2015.
2. "A Software Framework for the Global Optimization of Nonconvex Two-Stage Stochastic Programs," *AIChE Annual Meeting*, Nov. 2015.
1. "A Decomposition Strategy for a Class of of Nonconvex Two-Stage Stochastic Programs," *AIChE Annual Meeting*, Nov. 2014.

Poster

7. "Integrated Learning and Optimization," 3rd Annual Wisconsin Institute for Discovery Symposium, *University of Wisconsin-Madison*, Aug. 2020.
6. "Predict, then Smart Optimize with Stochastic Programming," *IPAM Workshop on Intersections between Control, Learning and Optimization*, Feb. 2020.
5. "A Stochastic Approximation Method for Chance-Constrained Nonlinear Programs," *AIChE Annual Meeting*, Nov. 2019.
4. "Optimization & Control of Chemical Process Systems Under Uncertainty," *AIChE Annual Meeting*, Nov. 2019.
3. "Stochastic Approximation for Chance-Constrained NLPs," Computing in Engineering Forum, *University of Wisconsin-Madison*, Sept. 2019.
2. "Stochastic Approximation for Chance-Constrained NLPs," *ICERM Workshop on Optimization of Systems Impacted by Rare, High-Impact Random Events*, June 2019.
1. "Optimizing Uncertain Systems with Reliability Requirements," 2nd Annual Wisconsin Institute for Discovery Symposium, *University of Wisconsin-Madison*, June 2019.

STUDENT RESEARCH PRESENTATIONS

External presentations made by students I have advised/mentored, including before joining VT. ★ = Undergraduate Advisee

Invited/Contributed

7. D. Sundararajan, "Machine Learning-Guided Undercover for Mixed-Integer Nonlinear Programs," *INFORMS Annual Meeting*, Nov. 2026.
6. J. Sanborn★, "A Machine Learning-Guided Heuristic for Mixed-Integer Nonlinear Programs," *IISE Annual Conference & Expo*, May 2026.
5. J.-C. Raymond-Bertrand, "Multistage Stochastic Dispatch of Mobile Power Sources for Enhanced Grid Resilience under Extreme Weather," *INFORMS Optimization Society Conference*, March 2026.
4. D. Sundararajan, "Machine Learning-Guided Undercover for Mixed-Integer Nonlinear Programs," *INFORMS Optimization Society Conference*, March 2026.
3. E. George, "A Generalizable End-to-End Learning Approach for Global Optimization of QCQPs," *25th International Symposium on Mathematical Programming*, July 2024.
2. E. George, "A Generalizable End-to-End Learning Approach for Global Optimization of QCQPs," *INFORMS Optimization Society Conference*, March 2024.
1. M. Goutham, "Optimal Power Flow for Resilient Electric Grids under Uncertain Hurricane Forecasts: A Multi-Stage Stochastic Programming Approach," *SIAM Conference on Optimization*, May 2023.

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4. J.-C. Raymond-Bertrand, "Multistage Stochastic Dispatch of Mobile Power Sources for Enhanced Grid Resilience under Extreme Weather," *LANL Grid Science Winter School and Conference*, Jan. 2027.
3. G. Esperanza★, "A Machine Learning-Guided Heuristic for Mixed-Integer Nonlinear Programs," ISE Undergraduate Research Symposium, *Virginia Tech*, May 2026.
2. A. S. Anchalwar and P. Rengasamy, "Deep Learning Surrogate Model for Property Prediction of Metal Organic Frameworks", *2025 CCI Workshop: AI for Science with Foundation Models, Virginia Tech*, May 2025.
1. M. Goutham, "Optimal Power Flow for Resilient Electric Grids under Uncertain Hurricane Forecasts: A Multi-Stage Stochastic Programming Approach," *LANL Grid Science Winter School and Conference*, Jan. 2023.